

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
LIDAM Energy Group, CORE, UCLouvain

Why this doctorate

I finish an M.Sc. in Economics at the University of Cologne this month, with a specialisation in energy economics and a thesis in applied microeconometrics. Energy economics has followed me through both degrees. What keeps me in it is that decarbonising our economies will be a monumental effort, one that reaches well past the capital it needs into how our societies work, and that almost all of it runs through instruments governments design. The rules of those instruments decide what actually gets built. I want to spend four years on that.

Why CORE

CORE describes itself as an interdisciplinary centre bringing operations research, economics and econometrics together, and it names doctoral training as one of its objectives. That is why I am writing to you and not to a department that does one of the three. My own work sits on the econometrics side, and what I do not yet have is graduate industrial organization and game theory at the level this project runs on. I have started without being asked to, working through Jehle and Reny in my own time. Everything I have written so far I wrote alone, and an Erasmus semester at Rotterdam School of Management taught me I work better otherwise. I like a room where nobody shares my background, and I like explaining what I am working on and hearing what other people are working on. That is most of what a research group is, and I want it next among thirty-five doctoral students and the visiting researchers you host.

What I would bring

My thesis, graded 1.0 and worth 30 of the degree's 120 ECTS, asked whether a national scholarship programme in Brazil raised formal-sector wages and for whom. I built the panel from administrative records, five million linked employer-employee observations across 558 microregions, 2005 to 2019, in R through BigQuery, and estimated Callaway and Sant'Anna (2021) group-time effects with doubly-robust inference across exposure bins. Low-exposure regions show a wage gain of 3.04 per cent, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response that two-way fixed effects reports with the opposite sign. That transfers here. Since the 2016 state aid framework member states have adopted technology-neutral, technology-specific and mixed auction designs at different dates, which is staggered adoption of a design rather than of a policy, and the AURES II database records the design and outcome of every renewable auction in the EU and UK from 2011 to 2021. I would estimate how joint bidding shifts the awarded technology portfolio and the rent per megawatt hour against not-yet-treated member states. Alongside the econometrics I have advanced microeconomics on strategic thinking and competition with Axel Ockenfels at 1.3, 1.0 in mathematics, and I have calibrated a model in MATLAB. I am free from October 2026 and my French is B1.

Thank you for considering my application. I would be glad to talk the project through whenever it suits you.

Referees

Prof. Dr. Pia Pinger, University of Cologne, chair of my master's thesis.
pia.pinger@uni-koeln.de

Jun.-Prof. Dr. Oliver Ruhnau, University of Cologne and EWI, examiner in *Energy Markets and Regulation*.
ruhnau@wiso.uni-koeln.de

Baha Khammari · b.e.khammari@gmail.com · +49 1520 9016956